

NAFISA BERLIANA INDAH PRATIWI

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EDUCATION

IPB University GPA: 3.42 M.Si. Statistics and Data Science	Jul 2021 - Aug 2024 Bogor, Indonesia
Universitas Sebelas Maret GPA: 3.16 S.Mat. Mathematics	Aug 2016 - April 2020 Surakarta, Indonesia
DQLab Bootcamp Data Analyst with Python and SQL Bootcamp Machine Learning & AI for Beginner	June 2024 - October 2024
International Language Program (ILP) TOEFL ITP Score: 553	May 2025

EXPERIENCE

Freelance Quantitative Analyst Remote	2021 - Present
- Conducted quantitative analysis with a focus on econometric modeling, complemented by statistical inference, exploratory data analysis, and machine learning techniques across diverse social research topics and policy briefs. - Created visualizations and dashboards using ggplot2 and Google Looker Studio to support data-driven insights. - Designed conceptual diagrams and flowcharts to map analytical workflows and research logic; proficient in Microsoft Visio and related tools. - Contributed to academic writing, including structuring papers, refining content, and improving language clarity for publication or thesis submission. - Designed survey instruments, selected appropriate methods, and performed end-to-end data analysis with emphasis on research validity.	
Supply Demand Staff PT.Sukanda Djaya	Sept 2025 - May 2026
- Managed warehouse records and operational documentation for rented warehouse activities, ensuring accuracy and completeness of Delivery Notes, Purchase Orders, and related transactions. - Reviewed import documentation and coordinated the goods arrival process, including document submission and SAP transaction processing for multiple brands.	
Instructor Assistant Quantitative Research Lab, Department of Economics, IPB University	Mar 2022 - Present
- Assisted participants during hands-on sessions on applied econometrics using RStudio, organized by the Quantitative Research Lab, Department of Economics. - Guided participants in implementing regression models, time series analysis, and diagnostic tests using R packages. - Clarified statistical concepts and helped troubleshoot common coding issues in R during guided exercises.	
Book Author	Dec 2023
- Contributed to the book Metode Kuantitatif dengan RStudio: Penerapan untuk Riset Manajemen dan Bisnis. - Authored the chapter on cluster analysis and contributed to additional sections on applied R-based methods.	
Teaching Assistant – Data Exploration Universitas Sebelas Maret (UNS), Indonesia	Jul 2020 – Dec 2020
- Provided academic support in Exploratory Data Analysis, guiding students through coursework and interpretation. - Assisted final-year students in the Soft Computing Research Group with thesis development and progress tracking.	

- Evaluated student interests and talents through interviews and academic records within the International Baccalaureate framework.
- Recommended curriculum adjustments to enhance alignment with student profiles and learning outcomes.

PROJECTS

City Branding Research

Jun 2025 - Present

Independent Project

- Designing a comprehensive city branding questionnaire covering key indicators such as cultural identity, economic competitiveness, stakeholder engagement, and digital presence.
- Developing Likert-scale survey instruments targeted at multiple stakeholder groups including government, academics, business owners, and cultural figures.
- Constructing an analytical framework grounded in city branding theory (e.g., place identity, heritage value, economic positioning).
- Currently collecting and analyzing stakeholder perception data alongside secondary sources, including Statistics Indonesia (BPS) and regional planning documents.
- Visualizing preliminary findings and documenting research progress using Quarto for iterative feedback and future policy application.

Smart City Index Research

Feb 2025 - June 2025

Independent Project

- Designed a tailored survey instrument to assess smart city dimensions, incorporating national and theoretical constructs.
- Implemented Analytic Hierarchy Process (AHP) with expert respondents to derive weighted scores across key pillars.
- Developed an original evaluation framework by synthesizing Kemenkominfo's national indicators with the Giffinger model.
- Collected and visualized secondary data from regional governments and Statistics Indonesia (BPS) to contextualize dimension-level scores.
- Produced cross-framework insights revealing structural alignment and divergence between national policy and global smart city theory.

Microplastic Perception Study

Dec 2024

Quantitative Analyst (Contributor)

Makassar, Indonesia

- Conducted behavioral clustering analysis using k-prototype algorithm on household survey data related to microplastic awareness and actions.
- Applied Silhouette Index for optimal cluster selection, identifying two distinct behavioral-geographic segments within the urban district.
- Implemented Random Forest-based feature importance analysis to interpret key drivers such as education level, environmental attitudes, and length of residence.
- Interpreted results in relation to geographic features (e.g., landfills, lakes) and identified socio-environmental disparities in engagement with plastic pollution.
- Contributed to research recommendations for geographically targeted public awareness campaigns and policy interventions.

UEQ-Based UX Analysis - User Experience Evaluation

Sept 2024 - Nov 2024

Independent Academic Project

- Supported a UX research project by analyzing website evaluation data using the standardized User Experience Questionnaire (UEQ), which measures six core dimensions: attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty.
- Applied non-parametric hypothesis testing (Kruskal-Wallis test) to identify significant differences in user experience ratings across multiple websites.
- Interpreted patterns in ordinal-scale responses and summarized key findings to assist in academic reporting and usability recommendations.

AWARDS & PRESENTATIONS

Winner – Call for Papers

2023

Sidang Pleno ISEI XXIII & Seminar Nasional

- Paper Title: Interkoneksi Pertumbuhan Ekonomi Antar Provinsi di Indonesia
- Selected as one of the best papers through national-level peer review and presented during the plenary session of the annual ISEI conference.

Presenter – International Academic Paper Presentation

2023

Sidang Pleno ISEI XXIII & Seminar Nasional

Presenter – International Academic Paper Presentation

2021

International Conference on Mathematics and Learning Research (ICOMER)

Presenter – International Academic Paper Presentation

2021

International Conference of Mathematics and Mathematics Education (ICMME)

Presenter – International Academic Paper Presentation

2020

International Conference on Mathematics: Education, Theory, and Application (ICMETA)

Presenter – International Academic Paper Presentation

2019

Ahmad Dahlan International Conference on Mathematics and Mathematics Education (ADINTERCOMME)

PUBLICATIONS

Nafisa Berliana Indah Pratiwi, Indahwati, F. Anwar, “[Village Potential Mapping: Comprehensive Cluster Analysis of Continuous and Categorical Variables with Missing Values and Outliers Dataset in Bogor, West Java, Indonesia](#)”, *Scientific Journal of Informatics*, 2024.

Nafisa Berliana Indah Pratiwi, D.R.S. Saputro, P. Widyaningsih, “[Fuzzy c-shells for clustering of data’s LQ45 in Indonesia based on earnings per share and price earning ratio](#)”, *Journal of Physics: Conference Series*, 2021.

Nafisa Berliana Indah Pratiwi, D.R.S. Saputro, “[Fuzzy c-shells clustering algorithm](#)”, *Journal of Physics: Conference Series*, 2020.

D.R.S. Saputro, P.H. Utomo, H. Prasetyo, **Nafisa Berliana Indah Pratiwi**, “[Evaluation of hybrid algorithms \(constraint base and bound and collapse\) in building Bayesian networks structures](#)”, *AIP Conference Proceedings*, 2022.

D.R.S. Saputro, N. Aulia, P.Y. Astuti, **Nafisa Berliana Indah Pratiwi**, S. Sutanto, “[Error correction models \(Case of data inflation in Indonesia\)](#)”, *AIP Conference Proceedings*, 2022.

D.R.S. Saputro, **Nafisa Berliana Indah Pratiwi**, R. Kusumawati, “[Logistic smooth transition autoregressive model parameter estimation using Gauss Newton](#)”, *AIP Conference Proceedings*, 2022.

D.R.S. Saputro, A. Susanti, **Nafisa Berliana Indah Pratiwi**, “[The handling of overdispersion on Poisson regression model with the generalized Poisson regression model](#)”, *AIP Conference Proceedings*, 2021.

D.R.S. Saputro, Y.K. Wardani, **Nafisa Berliana Indah Pratiwi**, P. Widyaningsih, “[Data simulation with Markov Chain Monte Carlo, Gibbs sampling, and Bayes \(beta-binomial\) methods as the parameter estimations of spatial bivariate probit regression model](#)”, *AIP Conference Proceedings*, 2021.

I.L. Shabrina, D.R.S. Saputro, **Nafisa Berliana Indah Pratiwi**, P. Pangadi, “[Liquid 45 modeling based on Markov chain and fuzzy linguistic summary using fuzzy stochastic](#)”, *AIP Conference Proceedings*, 2021.

SKILLS

Expertise: Econometrics, Statistical Inference, Exploratory Data Analysis, Survey Design, Data Visualization, Machine Learning, Clustering Analysis (e.g. k-means, k-modes, k-prototype, hierarchical, soft clustering), Time Series Analysis, Hypothesis Testing, Policy Analysis

Software: R (advanced), SPSS, Stata, Microsoft Office, Microsoft Visio, Google Looker Studio, SQL (basic), Quarto, LaTeX, Git, SAP

Languages: R, Python (basic), SQL (basic)